

論文リスト

1 Journal Articles

1. Y. Honda, O. Morikawa, S. Onoda and H. Suzuki,
“Lattice realization of the axial $U(1)$ non-invertible symmetry,”
Prog. Theor. Exp. Phys. **2024**, no.4, 043B04 (2024) [[arXiv:2401.01331](#) [hep-lat]].
2. N. Kan, O. Morikawa, Y. Nagoya and Hiroki Wada
“Erratum to: Higher-group structure in lattice Abelian gauge theory under instanton-sum modification,”
Eur. Phys. J. C **84**, no.1, 22 (2024).
3. M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
“Magnetic operators in 2D compact scalar field theories on the lattice,”
Prog. Theor. Exp. Phys. **2023**, no.7, 073B01 (2023) [[arXiv:2304.14815](#) [hep-lat]].
4. M. Abe, O. Morikawa and S. Onoda,
“Note on lattice description of generalized symmetries in $SU(N)/\mathbb{Z}_N$ gauge theories,”
Phys. Rev. D **108**, 014506 (2023) [[arXiv:2304.11813](#) [hep-th]].
5. M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
“Topology of $SU(N)$ lattice gauge theories coupled with $\mathbb{Z}_N$ 2-form gauge fields,”
J. High Energy Phys. **08**, 118 (2023) [[arXiv:2303.10977](#) [hep-lat]].
6. N. Kan, O. Morikawa, Y. Nagoya and H. Wada,
“Higher-group structure in lattice Abelian gauge theory under instanton-sum modification,”
Eur. Phys. J. C **83**, no.6, 481 (2023) [[arXiv:2302.13466](#) [hep-th]].
7. O. Morikawa, H. Wada and S. Yamaguchi,
“Phase structure of linear quiver gauge theories from anomaly matching,”
Phys. Rev. D **107**, 045020 (2023) [[arXiv:2211.12079](#) [hep-th]].
8. M. Abe, O. Morikawa and H. Suzuki,
“Fractional topological charge in lattice Abelian gauge theory,”
Prog. Theor. Exp. Phys. **2023**, no.2, 023B03 (2023) [[arXiv:2210.12967](#) [hep-th]].
9. H. Makino, O. Morikawa and H. Suzuki,
“Gradient flow and the Wilsonian renormalization group flow,”
Prog. Theor. Exp. Phys. **2021**, no.9, 099201 (2021).
10. M. Ashie, O. Morikawa, H. Suzuki and H. Takaura,
“More on the infrared renormalon in $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$,”
Prog. Theor. Exp. Phys. **2020**, no.9, 093B02 (2020) [[arXiv:2005.07407](#) [hep-th]].

11. O. Morikawa and H. Takaura,
“Identification of perturbative ambiguity canceled against bion,”
Phys. Lett. B **807**, 135570 (2020) [arXiv:2003.04759 [hep-th]].
12. K. Ishikawa, O. Morikawa, K. Shibata and H. Suzuki,
“Vacuum energy of the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$ in the $1/N$ expansion,”
Prog. Theor. Exp. Phys. **2020**, no.6, 063B02 (2020) [arXiv:2001.07302 [hep-th]].
13. K. Ishikawa, O. Morikawa, A. Nakayama, K. Shibata, H. Suzuki and H. Takaura,
“Infrared renormalon in the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$,”
Prog. Theor. Exp. Phys. **2020**, no.2, 023B10 (2020) [arXiv:1908.00373 [hep-th]].
14. M. Ashie, O. Morikawa, H. Suzuki, H. Takaura and K. Takeuchi,
“Infrared renormalon in $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$,”
Prog. Theor. Exp. Phys. **2020**, no.2, 023B01 (2020) [arXiv:1909.05489 [hep-th]].
15. K. Ishikawa, O. Morikawa, K. Shibata, H. Suzuki and H. Takaura,
“Renormalon structure in compactified spacetime,”
Prog. Theor. Exp. Phys. **2020**, no.1, 013B01 (2020) [arXiv:1909.09579 [hep-th]].
16. O. Morikawa,
“Continuum limit in numerical simulations of the $\mathcal{N} = 2$ Landau–Ginzburg model,”
Prog. Theor. Exp. Phys. **2019**, no.10, 103B03 (2019) [arXiv:1906.00653 [hep-lat]].
17. O. Morikawa,
“Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model with two superfields,”
J. High Energy Phys. **12**, 045 (2018) [arXiv:1810.02519 [hep-lat]].
18. A. Kasai, O. Morikawa and H. Suzuki,
“Gradient flow representation of the four-dimensional $\mathcal{N} = 2$ super Yang–Mills supercurrent,”
Prog. Theor. Exp. Phys. **2018**, no.11, 113B02 (2018) [arXiv:1808.07300 [hep-lat]].
19. O. Morikawa and H. Suzuki,
“Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model,”
Prog. Theor. Exp. Phys. **2018**, no.8, 083B05 (2018) [arXiv:1805.10735 [hep-lat]].
20. O. Morikawa and H. Suzuki,
“Axial $U(1)$ anomaly in a gravitational field via the gradient flow,”
Prog. Theor. Exp. Phys. **2018**, no.7, 073B02 (2018) [arXiv:1803.04132 [hep-th]].
21. H. Makino, O. Morikawa and H. Suzuki,
“Gradient flow and the Wilsonian renormalization group flow,”
Prog. Theor. Exp. Phys. **2018**, no.5, 053B02 (2018) [arXiv:1802.07897 [hep-th]].
22. H. Makino, O. Morikawa and H. Suzuki,
“One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator,”
Prog. Theor. Exp. Phys. **2017**, no.6, 063B08 (2017) [arXiv:1704.04862 [hep-lat]].

23. H. Makino and O. Morikawa,
“Lorentz symmetry violation in the fermion number anomaly with the chiral overlap operator,”
Prog. Theor. Exp. Phys. **2016**, no.12, 123B06 (2016) [arXiv:1609.08376 [hep-lat]].

2 Preprints

1. O. Morikawa, S. Onoda and H. Suzuki,
“Yet another lattice formulation of 2D $U(1)$ chiral gauge theory via bosonization,”
arXiv:2403.03420 [hep-lat].

3 Thesis

1. O. Morikawa
“Numerical study of infrared criticality of the supersymmetric Landau–Ginzburg model,”
Ph.D. Thesis, Department of Physics, Kyushu University, 2021.
DOI: 10.13140/RG.2.2.36216.42246
HDL: 2324/4474929

4 Proceedings

1. M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
“Lattice construction of mixed 't Hooft anomaly with higher-form symmetry,”
Proc. Sci. **LATTICE2023**, 361 (2024) [arXiv:2401.00495 [hep-lat]].
2. M. Abe, N. Kan, O. Morikawa, Y. Nagoya, S. Onoda and H. Wada,
“Higher-group symmetry in lattice gauge theories with restricted topological sectors,”
Proc. Sci. **LATTICE2023**, 365 (2024) [arXiv:2310.01787 [hep-lat]].
3. O. Morikawa,
“Numerical study of ADE-type $\mathcal{N} = 2$ Landau–Ginzburg models,”
Proc. Sci. **LATTICE2019**, 145 (2020) [arXiv:1908.03411 [hep-lat]].
4. H. Makino, O. Morikawa and H. Suzuki,
“One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator,”
EPJ Web Conf. **175**, 11013 (2018) [arXiv:1710.00536 [hep-lat]].